

AI-BASED SMART BLOOD DONOR FINDER

Project Proposal & IEEE 2025–2026 Literature Benchmark Sheet

Using Intelligent Matching, Location-Based Services, and Blockchain Ledger

EXECUTIVE SUMMARY:

During critical medical emergencies, finding compatible blood donors quickly is a life-or-death challenge. Conventional methods relying on manual phone calls, WhatsApp broadcasting, or static directory listings suffer from low response rates, privacy concerns, lack of distance accuracy, and zero verification.

This project introduces an **AI-Powered, Privacy-Preserving Smart Blood Donor Finder**. Rather than performing simple database filtering, the system uses a **Multi-Factor Weighted AI Ranking Algorithm**, a **Bilingual (English / Tamil) NLP AI Assistant**, an **Anonymized Geolocation Privacy Shield**, and a **Cryptographic Blockchain Ledger** to ensure transparent, secure, and rapid donor-to-patient connections.

1. IEEE 2025–2026 Literature Benchmark vs. Our System

Recent IEEE publications in 2025 and 2026 focus on isolated machine learning predictions (e.g., Random Forest for donor retention) or complex enterprise B2B hospital supply chains (Hyperledger). They lack a unified real-time P2P matching engine with regional language NLP and zero-knowledge location privacy.

Feature / Criteria	IEEE 2025–2026 Literature Baseline	Our Proposed System (2026+ Novel Points)
AI Matching Engine	Single-variable ML (e.g. static donor retention score) or simple city filtering.	Unified Real-Time Multi-Factor AI Engine Scores 0–100% using ABO matrix, Haversine distance, availability, eligibility, and urgency.
Regional Language NLP	Monolingual English forms or basic voice input.	Bilingual English & Tamil NLP Parser Parses natural text/voice (e.g., "B+ 2" to auto-fill requests.
Geolocation Privacy	Exposes exact GPS lat/long coordinates or public phone directory.	Dynamic Anonymized Radius Shielding Shows "Approx 2.3 KM away". Contact details unlocked strictly after 2-way consent .
Blockchain Technology	Heavy enterprise B2B Hyperledger (Hospital to Bank only).	Lightweight P2P Micro-Blockchain Ledger Instant SHA-256 hash verification & digital proof-of-donation certificates.
Availability Engine	Binary Active / Inactive status.	Tri-State Availability Engine Available Busy Temporarily Unavailable.
Emergency Urgency	Standard FIFO Queue.	Real-Time Dynamic Urgency Weighting Critical / Urgent / Standard priority scaling.

2. Detailed Breakdown of the 8 Core Modules

• **Module 1: User Authentication & Security Module**

Features Role-Based Access Control (RBAC) for Donors, Requesters, and Admins. Integrates dynamic Privacy Address Shielding so a donor's exact home location and phone number are hidden until consent is given.

- **Module 2: Smart Matching & Proximity Engine**

Uses a Multi-Factor AI Weighted Scoring Algorithm: $\text{Match Score} = (w1 * \text{Compatibility}) + (w2 * \text{Haversine Distance}) + (w3 * \text{Availability}) + (w4 * \text{Eligibility Gap}) + (w5 * \text{Urgency})$. Includes interactive Leaflet map radius visualization.

- **Module 3: AI Assistant & Verification Module**

Features a bilingual (English & Tamil) NLP request parser that automatically extracts Blood Group, Hospital Location, Units, and Urgency from conversational user input. Includes a smart donor health eligibility validator (> 90 days gap check).

- **Module 4: Blockchain Ledger Module (Major Novelty ■)**

Implements an immutable SHA-256 cryptographic hash-linked ledger for all completed blood donations. Generates tamper-proof digital certificates to eliminate fake donation claims.

- **Module 5: Administrative & Analytics Module**

Executive Command Dashboard providing real-time KPI metrics, donor availability by blood group, critical emergency dispatch monitoring, and a live blockchain audit log inspector.

- **Module 6: Donors Module**

Provides a Tri-State Availability Switcher (■ Available / ■ Busy / ■ Temporarily Unavailable), an instant emergency request notification panel with [Accept] / [Decline] buttons, and a digital certificate wallet.

- **Module 7: Requesters Module (Patients & Hospitals)**

Features a 3-step Emergency Blood Request Creator, real-time ranked donor candidate dashboard with match percentage breakdowns, and a live 4-stage request tracker.

- **Module 8: Language Module (English & ■■■■■)**

Native i18n localization engine allowing seamless 1-click toggling between English and Tamil across all forms, buttons, alerts, AI chatbot prompts, and certificates.

3. 1-Minute Pitch Script for Client / Academic Review

CLIENT / PROFESSOR PRESENTATION SCRIPT:

*"Standard blood donor apps simply list phone numbers on a screen. Our project is an **AI-Based Smart Blood Donor Finder**.*

*When an emergency occurs, our AI calculates a **0–100% Match Score** using distance, blood compatibility, donor eligibility, and emergency urgency. Patients can request blood in **Tamil or English** using our AI Chatbot.*

*To protect donor privacy, exact locations are hidden until a donor accepts. Finally, every verified donation is recorded on an immutable **Blockchain Ledger** to issue digital proof certificates. This makes our system far superior to existing IEEE papers."*